- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

12 A box of mass m kg slides down a rough slope inclined at 15° to the horizontal. The coefficient of friction between the box and the slope is 0.4. The box has an initial velocity of $1.2\,\mathrm{m\,s^{-1}}$ down the slope.

Calculate the distance the box travels before coming to rest. **[7]**